

FORD VEV

Proving Ground Operational System

Aplicativo de Coleta de Dados - PWA

Documentacao Tecnica - ABNT NBR 14724

Gerado em 08/06/2026

Aplicação single-page (SPA) em JavaScript puro, sem frameworks, que funciona como PWA instalável para coleta de dados em campo no Campo de Provas Ford Tatuí.

1 Screens

The app has 3 main screens navigated via a bottom tab bar:

Screen	ID	Purpose
Central	`screen-home`	Shift management, quick actions, bento m
Telemetria	`screen-telemetry`	GPS route logging, speed telemetry
Histórico	`screen-history`	Report feed (laudos, trips, shifts), PDF

2 User Flow

```
Login (@ford.com) -> Home Screen -> Iniciar Turno (choose project/vehicle)
+-- Roteiro R389 (lap-by-lap checklist)
+-- Desaceleração 16 Laps (deceleration protocol)
+-- Protocolo de Frenagem (braking test, 8 checkpoints)
+-- Laudo Técnico (issue report with photo, severity)
+-- Abastecimento (fuel refueling log)
+-- Copiloto GPS (telemetry/route recording)
+-- Encerrar Turno (final km, liters, fuel station)
+-- Data saved to Firebase (Firestore + RTDB)
```

3 Core Modules

3.1 `app.js` (~4,000 lines)

Main application logic: Firebase init, turno state, GPS tracking, UI event handlers, PDF generation, data persistence, modal management. Contains the app object with all state variables (fotos, videosFiles, checkins, rastrosGps, etc.).

3.2 `turno-engine.js` (~3,159 lines)

Two engines:

- PreCadastroEngine - localStorage-based CRUD for operators and fuel stations
- TurnoEngine - Full shift lifecycle: iniciar(), adicionarAbastecimento(), sincronizarComApp(), encerrar(). Saves to both localStorage (active turno cache) and Firebase RTDB (vev_turnos_ativos)

3.3 `auth-security.js` (~1,160 lines)

- Firebase Auth with email/password, LOCAL persistence
- @ford.com domain auto-approval
- 3-attempt logout (5 min cooldown)

- EmailJS integration for approval workflows
- Security audit logs to Firestore (vev_logs_seguranca)
- Operator auto-registration on login

3.4 `laudo-ia.js` (~1,069 lines)

Technical issue (laudo) creation with:

- Photo capture (compressed to 800px max)
- Severity selection (Leve/Moderado/Grave/Crítico)
- System affected (motor, transmissão, freios, suspensão, elétrica, etc.)
- Gemini API text refinement
- PDF generation via jsPDF with autoTable (5-page template)

3.5 `analytics.js` (~775 lines)

- Trip analytics (export to Google Maps URL)
- Speed/ RPM summary, distance breakdown
- Per-vehicle FEG (fuel efficiency) trends
- Battery voltage tracking

3.6 `dados-mestres.js` (219 lines)

Master data registry with Firestore sync:

- PROJETOS: Active projects (EET Ranger 2024, Ranger VoCF, Testes Especiais, 1M Mile)
- TESTES_PISTA: Track test types (Durabilidade, Especiais × Interno/Externo)
- Methods: getProjetos(), getTestesPista(), getVeiculos(), getPostos()

3.7 `forms-engine.js` (123 lines)

Dynamic modal for project-specific Microsoft Forms links. Displays a list of form buttons per project, opening forms in new tabs.

3.8 `notificacoes-engine.js` (71 lines)

Firestore-based notification system: turnoEncerrado(), issueCritico(), kmExcedido(), with real-time listener for coordinator panel.

4 Modals (17+)

Modal	Trigger	Purpose
`modal-inicio-turno`	"Iniciar Turno"	Project, ambiente, vehicle, initial km
`modal-turno`	"Encerrar Turno"	Final km, liters, fuel station, observat

`modal-roteiro`	"Roteiro R389"	Lap-by-lap checklist (R389 protocol)
`modal-desaceleracao`	"Desaceleração"	16-lap deceleration protocol
`modal-frenagem`	"Frenagem"	Braking test protocol (8 checkpoints)
`modal-laudo`	"Laudo Técnico"	Issue report form
`modal-laudos-pendentes`	Home indicator	View pending/open laudos
`modal-gps`	"Copiloto GPS"	Real-time GPS track recording
`modal-utilidades`	"Utilidades"	Quick tools/miscellaneous
`modal-analytics`	"Analytics"	Driver performance charts
`modal-aprovacoes`	Admin	Approval panel for new users
`modal-configuracoes-sistema`	Admin	System settings, log viewer
`modal-gerenciar-forms`	Admin	Form management
`modal-historico-turnos`	"Histórico"	Past shift history
`modal-historico-performance`	"Performance"	Performance history
`modal-login`	App start	Firebase Auth login
`modal-recuperacao`	Login screen	Password recovery

5 Telemetry (GPS)

The Copiloto GPS system records:

- GPS coordinates (batched to RTDB at vev_gps_batch/)
- Instant speed -> speed bands (<80, 80-160, 160-230, >230 km/h)
- Distance traveled, average speed, max speed
- Chart.js real-time speed graph
- Route export to Google Maps URL

6 PDF Generation

Generated via jsPDF + autoTable + html2canvas:

- Laudo template: 5 pages - header, vehicle data, occurrence description, photo page, digital signature
- Route report: Telemetry stats, speed distribution, map snapshot

7 PWA Features

- Service Worker: ford-vev-cache-v9, network-first with offline fallback
- Precached: 14 assets (all JS modules, index.html, offline.html)
- Installable: display: standalone, maskable icons, theme color #003399
- Offline page: offline.html with "Você está offline" message and retry button

8 Build (Vite)

```
// vite.config.js highlights
```

- Build output: dist/
- JS minification: esbuild (drops console.log + debugger)
- Static JS files copied verbatim (not bundled) for Firebase compat SDK
- Assets cached 1 year (immutable) in firebase.json

9 Key Routes (app.js)

Function	Purpose	
`app.iniciarTurno()`	Start shift, save to localStorage + RTDB	
`app.encerrarTurno()`	End shift, save to Firestore collection	
`app.adicionarAbastecimento()`	Log fuel refueling	
`app.gerarRelatorioRodagem()`	Generate GPS route PDF	
`app.salvarRastroGps()`	Batch-save GPS points	
`app.abrirModal()` / `app.fecharModal()`	Modal management	
`switchScreen('home'`	'telemetry'	'histo